

6.1. Radar configuration (0x200)

The ARS 404-21 and 408-21 can be configured via message **RadarCfg** (0x200). The parameters can be changed individually or in combinations. For each parameter, the message contains a validity bit (e.g. parameter **RadarCfg_MaxDistance** with **RadarCfg_MaxDistance_valid**). If the validity bit is set to **valid** (0x1), the corresponding parameter will be updated in the ARS, otherwise it is ignored.

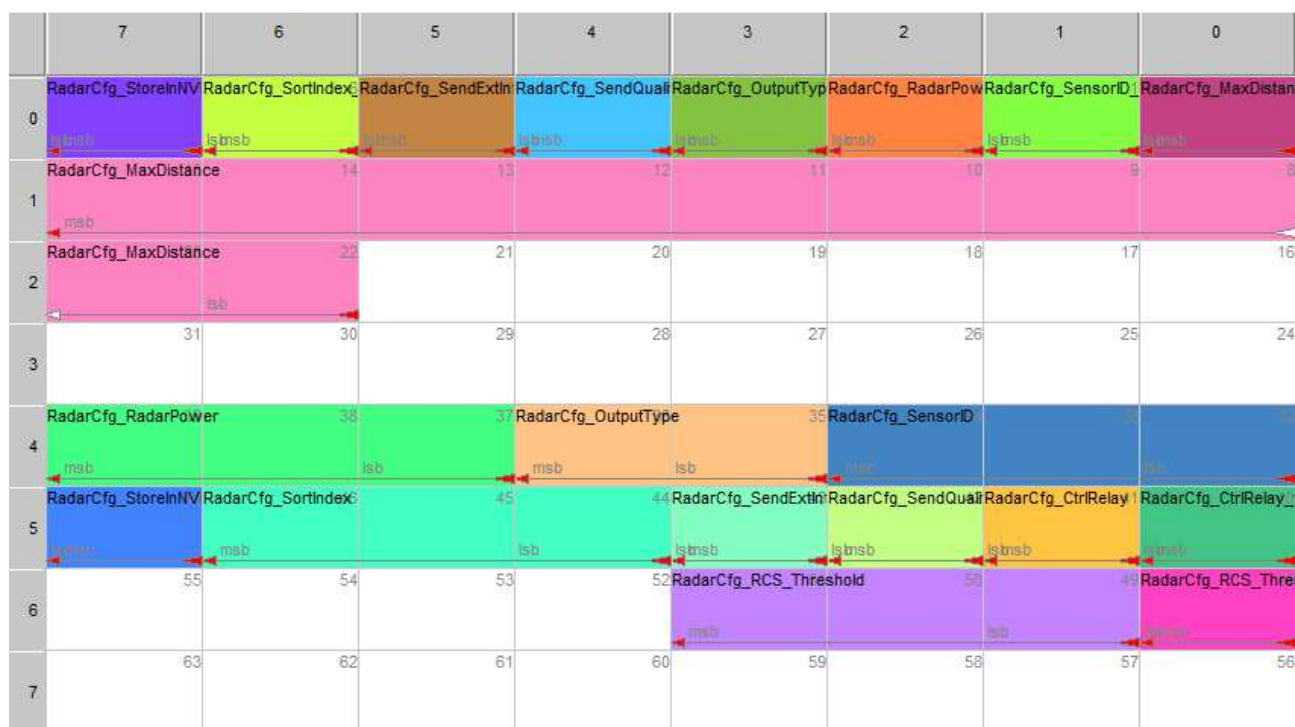


Figure 4: RadarCfg- message layout (0x200)

Table 2: RadarCfg – message content (0x200)

Signal	Start	Len	Min	Max	Res	Unit
RadarCfg_MaxDistance_valid	0	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SensorID_valid	1	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_RadarPower_valid	2	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_OutputType_valid	3	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SendQuality_valid	4	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_SendExtInfo_valid	5	1	0	1	1	0x0: invalid 0x1: valid

RadarCfg_SortIndex_valid	6	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_StoreInNVM_valid	7	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_MaxDistance	22	10	0	2046	2	m
RadarCfg_SensorID	32	3	0	7	1	Sensor ID 0x0 to 0x7
RadarCfg_OutputType	35	2	0	2	1	0x0: none "stand by" 0x1: send objects 0x2: send clusters 0x3: reserved – don't use it
RadarCfg_RadarPower	37	3	0	7	1	0x0: Standard 0x1: -3dB Tx gain 0x2: -6dB Tx gain 0x3: -9dB Tx gain
RadarCfg_CtrlRelay_valid	40	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_CtrlRelay	41	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_SendQuality	42	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_SendExtInfo	43	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_SortIndex	44	3	0	7	1	0x0: no sorting 0x1: sorted by range 0x2: sorted by RCS
RadarCfg_StoreInNVM	47	1	0	1	1	0x0: inactive 0x1: active
RadarCfg_RCS_Threshold_valid	48	1	0	1	1	0x0: invalid 0x1: valid
RadarCfg_RCS_Threshold	49	3	0	7	1	0x0: Standard 0x1: High sensitivity

Table 3: RadarCfg – signal description (0x200)

Start	Signal	Description
32	RadarCfg_SensorID	Sensor ID 0 – 7 (see also section 4 for further explanation)
1	RadarCfg_SensorID_valid	Allows change of sensor ID if true

22	RadarCfg_MaxDistance	Maximum distance of far scan (near scan maximum distance is set proportionally to half of the far scan). The maximum distance will also change the range resolution proportionally. For example, 200 m maximum distance will lead to: - far scan 200 m with 1.79 m range resolution, - near scan 100 m with 0.42 m range resolution. ARS408: Standard Range: 196 m – 260 m Extended Range: 196 m – 1200 m ARS404: Standard Range: 150 m – 190 m Extended Range: 90 m – 1000 m
0	RadarCfg_MaxDistance_valid	Allows change of maximum distance if true
37	RadarCfg_RadarPower	Configures the transmitted radar power (Tx attenuation). The output RCS of cluster and objects will be compensated for this attenuation. Reducing the output power can improve detection in case of close range scenarios or inside rooms.
2	RadarCfg_RadarPower_valid	Allows change of radar output power if true
35	RadarCfg_OutputType	Configures the data output to Clusters (0x2) or Objects (0x1) or even to "Stand By" (0x0) where only the IDs 0x700 and 0x201 are sent from the radar once a sec.. Please don't use (0x3) which is reserved.
3	RadarCfg_OutputType_valid	Allows the change of the output type if true
42	RadarCfg_SendQuality	Cluster or object quality information (message 0x60C or 0x702) is sent if true
4	RadarCfg_SendQuality_valid	Allows change of quality option if true
43	RadarCfg_SendExtInfo	Extended information (message 0x60D) is sent for objects if true (if clusters are selected as output type this value is ignored)
5	RadarCfg_SendExtInfo_valid	Allows change of extended information option if true
44	RadarCfg_SortIndex	Selects the sorting index for the object list (ignored for clusters as they are always sorted by range)
6	RadarCfg_SortIndex_valid	Allows change of sorting index if true
41	RadarCfg_CtrlRelay	Relay control message (0x8) is sent if true and the collision detection is activated
40	RadarCfg_CtrlRelay_valid	Allows change of relay control output if true
49	RadarCfg_RCS_Threshold	Sets the sensitivity of the cluster detection to standard (0x0) or high sensitivity (0x1)
48	RadarCfg_RCS_Threshold_valid	Allows change of RCS threshold option if true

47	RadarCfg_StoreInNVM	Stores the current configuration to non-volatile memory to be read and set at sensor startup.
7	RadarCfg_StoreInNVM_valid	Allows storing to non-volatile memory if true